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Wire harness

Abstract

A wire harness including; a wire harness main body including an electric wire and an exterior tube that covers an outer circumferential surface of the electric wire; a path restrictor that is attached to an outer circumferential surface of the exterior tube and restricts a path of the wire harness; and an attachment attached to an outer circumferential surface of a portion in a lengthwise direction of the path restrictor.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to a wire harness.
(2) Conventionally, a wire harness that includes: a wire harness main body that includes an electric wire member and an exterior member that covers the electric wire member; and a path restricting member that is attached to an outer circumferential surface of the exterior member and is configured to restrict the path of the wire harness main body, has been known as a wire harness for a vehicle (for example, see JP 2013-55760A).

SUMMARY

(3) Incidentally, in the above-described wire harness, it is desired to improve the connection reliability of the path restricting member with respect to another member such as the exterior

member.

(4) An exemplary aspect of the disclosure provides a wire harness capable of improving the connection reliability between the path restricting member and another member.

(5) The wire harness of the present disclosure includes: a wire harness main body including an electric wire and an exterior tube that covers an outer circumferential surface of the electric wire; a path restrictor that is attached to an outer circumferential surface of the exterior tube and restricts a path of the wire harness; and an attachment attached to an outer circumferential surface of a portion in a lengthwise direction of the path restrictor, wherein: the path restrictor includes a first main body that covers a portion of the outer circumferential surface of the exterior tube and an insertion port that is open in a direction orthogonal to a lengthwise direction of the first main body and extends over an entire length in the lengthwise direction of the first main body, the path restrictor has a connection connected to the attachment, the attachment has a cover that covers an outer circumferential surface of the connection, and the cover has a mark indicating a necessary minimum insertion amount of the path restrictor into the cover in a lengthwise direction of the wire harness main body.

(6) According to the wire harness of the present disclosure, an effect of being able to improve the reliability of connection between the path restricting member and other members is exhibited.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram showing a wire harness of an embodiment.

(2) FIG. 2 is a schematic perspective view showing a wire harness of an embodiment.

(3) FIG. 3 is a schematic exploded perspective view showing a wire harness of an embodiment.

(4) FIG. 4 is a schematic transverse cross-sectional view showing a wire harness of an embodiment.

(5) FIG. 5 is a schematic front view showing a wire harness of an embodiment.

(6) FIG. 6 is a schematic cross-sectional view showing a wire harness of an embodiment.

(7) FIG. 7 is a schematic perspective view showing a wire harness of a modified example.

(8) FIG. 8 is a schematic cross-sectional view showing a wire harness of a modified example.

DETAILED DESCRIPTION OF EMBODIMENTS

Description of Embodiments of Present Disclosure

(9) First, embodiments of the present disclosure will be listed and described.

(10) [1] The wire harness of the present disclosure includes: a wire harness main body including an electric wire member and an exterior member that covers an outer circumferential surface of the electric wire member; a path restricting member that is attached to an outer circumferential surface of the exterior member and restricts a path of the wire harness; and an attachment member attached to an outer circumferential surface of a portion in a lengthwise direction of the path restricting member, in which the path restricting member includes a first main body portion that covers a portion of the outer circumferential surface of the exterior member and an insertion port that is open in a direction orthogonal to a lengthwise direction of the first main body portion and extends over the entire length in the lengthwise direction of the first main body portion, the path restricting member has a connection portion connected to the attachment member, the attachment member has a covering portion that covers an outer circumferential surface of the connection portion, and the covering portion has a mark indicating a necessary minimum insertion amount of the path restricting member into the covering portion in a lengthwise direction of the wire harness main body.

(11) According to this configuration, the connection portion of the path restricting member and the covering portion of the attachment member are provided so as to overlap each other in the

lengthwise direction of the wire harness main body. At this time, the mark indicating the necessary minimum insertion amount of the path restricting member into the covering portion is provided on the covering portion that covers the outer circumferential surface of the connecting portion. For this reason, during the task of inserting the path restricting member into the covering portion, it is possible to easily confirm the position corresponding to the minimum insertion amount of the path restricting member into the covering portion based on the mark. Accordingly, by inserting the path restricting member into the covering portion while confirming the mark, the path restricting member can be suitably inserted into the covering portion by a necessary minimum insertion amount or more. As a result, it is possible to suitably suppress detachment of the path restricting member from the covering portion. As a result, it is possible to improve the connection reliability between the path restricting member and the attachment member. Note that, for example, an uneven structure, engraving, printing, or the like can be used as the mark.

(12) [2] It is preferable that the covering portion has a restricting wall that is engageable with a first end surface in the lengthwise direction of the path restricting member, in the lengthwise direction of the wire harness main body. According to this configuration, when the first end surface of the path restricting member is engaged with the restricting wall, relative movement of the path restricting member with respect to the attachment member is suppressed in the lengthwise direction of the wire harness body. For this reason, when the first end surface is engaged with the restricting wall, it is possible to suppress insertion of the path restricting member farther than the restricting wall of the attachment member. In other words, the restricting wall can function as a mark indicating the maximum insertion amount of the path restricting member into the covering portion in the lengthwise direction of the wire harness main body. As a result, the path restricting member can be inserted into the covering portion using the mark and the restricting wall as guides. According to this, by inserting the path restricting member into the covering portion such that the first end surface is positioned between the mark and the restricting wall, the path restricting member can be suitably inserted into the covering portion up to a desired position. As a result, it is possible to improve the efficiency of the task of inserting the path restricting member into the covering portion, and therefore it is possible to improve the assembly workability of the wire harness.

(13) [3] It is preferable that the restricting wall is provided at one end portion in a lengthwise direction of the covering portion, the covering portion has a second end surface of an end portion on a side opposite to the restricting wall out of the two end portions in the lengthwise direction of the covering portion, the mark is provided spaced apart from the second end surface by a first distance in the lengthwise direction of the wire harness main body, and the first distance is a distance corresponding to the minimum insertion amount. According to this configuration, the mark is provided spaced apart by the first distance corresponding to the minimum insertion amount from the second end surface provided on the side opposite to the regulating wall in the lengthwise direction of the covering portion. Based on such a mark and the restricting wall, it is possible to easily confirm the positions corresponding to the minimum insertion amount and the maximum insertion amount of the path restriction member with respect to the covering portion.

(14) [4] It is preferable that the covering portion is formed in a ring shape that encloses, in a circumferential direction, the entirety of the outer circumferential surface of the connection portion and the entirety of the outer circumferential surface of the exterior member at a portion where the connection portion is attached, and the covering portion has a second main body portion that covers a portion of the outer circumferential surface of the first main body portion, and a lid portion that is coupled to the second main body portion. According to this configuration, since the covering portion is ring-shaped and the covering portion is divided into the second main body portion and the lid portion, the attachment member including the covering portion can be retrofitted to the path restricting member and the exterior member. As a result, the assembly workability of the wire harness can be improved.

(15) [5] It is preferable that the mark is provided on an inner surface of the second main body portion. According to this configuration, since the mark is provided on the inner surface of the second main body portion, when inserting the path restricting member into the second main body portion, the position of the mark can be easily confirmed visually. Furthermore, it is easy to visually confirm the positional relationship between the mark and the first end surface of the path restricting member. As a result, the workability when inserting the path restricting member into the covering portion can be improved, and the path restricting member can be preferably inserted into the covering portion by the necessary minimum insertion amount or more.

(16) [6] It is preferable that the mark extends over the entirety of a circumferential surface of the second main body portion in a circumferential direction of the covering portion. According to this configuration, since the mark is formed in a wide range in the circumferential direction of the covering portion, it is easier to visually confirm the position of the mark.

(17) [7] It is preferable that the mark has a recess that is recessed from the inner surface of the second main body portion, and the recess is a recess that does not pass through the second main body portion in a radial direction of the covering portion. According to this configuration, since the recess serving as the mark does not pass through the second main body portion in the radial direction, the recess can be formed in a wide range in the circumferential direction of the covering portion. Also, since the recess is formed so as to be recessed from the inner surface of the second main body portion, it is possible to suitably suppress a case in which the recess hinders insertion of the path restricting member into the second main body portion.

(18) [8] It is preferable that the second main body portion has a third end surface and a fourth end surface, which are both end surfaces in a circumferential direction of the second main body portion, and the mark is provided on at least one of the third main body portion and the fourth main body portion. According to this configuration, the mark is provided on at least one of the third end surface and the fourth end surface of the lid portion. For this reason, even after the path restricting member is inserted into the second main body, the mark can be visually confirmed. This makes it possible to easily confirm the positional relationship between the first end surface of the path restricting member and the mark even after the path restricting member is inserted into the second main body portion.

(19) [9] It is preferable that when the path restricting member is a first path restricting member, the attachment member is a second path restricting member that is attached to an outer circumferential surface of the exterior member and restricts the path of the wire harness main body, the connection portion is provided at an end portion in a lengthwise direction of the first path restricting member, the covering portion is provided at an end portion in a lengthwise direction of the second path restricting member, the first path restricting member restricts a path of a straight section, which is a portion with a linear shape in the path of the wire harness main body, and the second path restricting member restricts a path of a bent section, which is a portion with a bent shape in the path of the wire harness main body. According to this configuration, the path of the straight section is restricted by the first path restricting member, and the path of the bent section is restricted by the second path restricting member. As a result, it is possible to suppress deviation of the path of the straight section and the path of the bent section from desired paths.

Detailed Description of Embodiments of the Present Disclosure

(20) A specific example of the wire harness of the present disclosure will be described below with reference to the drawings. In each drawing, some configurations may be exaggerated or simplified for convenience of description. Also, the dimensional ratio of each portion may differ in each drawing. “Orthogonal”, “parallel” and “entire length” in this specification include not only the case of being strictly orthogonal, parallel and the entire length, but also the case of being approximately orthogonal, parallel, and the entire length within a range where the effects in this embodiment are exhibited. In the present specification, the term “equal” includes not only the case of being exactly equal, but also the case where there is a slight difference between objects being compared due to

the influence of a dimensional tolerance or the like. Also, the term “tubular shape” used in the description of this specification includes not only a shape obtained by forming a circumferential wall continuously over the entire length in the circumferential direction, but also a shape in which a plurality of parts are combined to form a tubular shape, and a shape having a notch or the like in a portion in the circumferential direction, like a C shape. Note that “tubular” shapes include, but are not limited to, circles, ovals, and polygons with pointed or rounded corners. Also, the term “ring-shaped” used in the description of this specification may refer to any structure that forms a loop, a continuous shape without ends, as well as a structure that is generally loop-shaped with a gap, such as a C shape. Note that “ring-shaped” shapes include, but are not limited to, circles, ovals, and polygons with pointed or rounded corners. Also, the term “opposing” in this specification refers to surfaces or members being located in front of each other, and includes not only a case where the surfaces or members are located completely in front of each other, but also a case where the surfaces or members are partially located in front of each other. Also, “opposing” in this specification includes both a case where a member different from two portions is interposed between the two portions, and a case where nothing is interposed between the two portions. Note that the present disclosure is not limited to these examples, but is indicated by the scope of the claims, and is intended to include all modifications that fall within the scope and meaning equivalent to the scope of the claims.

(21) Overall Configuration of Wire Harness 10

(22) A wire harness **10** shown in FIG. **1** is to be mounted in a vehicle **V** such as a hybrid vehicle or an electric vehicle, for example. The wire harness **10** electrically connects two or more on-board devices to each other. The on-board devices are electric devices mounted in the vehicle **V**. The wire harness **10** electrically connects, for example, an inverter **M1** that is installed in a front portion of the vehicle **V** and a high-voltage battery **M2** that is installed rearward of the inverter **M1** in the vehicle **V**, to each other. The wire harness **10** is formed in an elongated shape so as to extend in a front-rear direction of the vehicle **V**, for example. The wire harness **10** is routed in the vehicle **V** so that, for example, an intermediate portion in the lengthwise direction of the wire harness **10** passes outside the vehicle interior such as under the floor of the vehicle **V**.

(23) The inverter **M1** is, for example, connected to a motor (not shown) for driving wheels, which serves as a power source when the vehicle travels. The inverter **M1** generates AC power from the DC power of the high-voltage battery **M2**, and supplies the AC power to the motor. The high-voltage battery **M2** is, for example, a battery that is capable of supplying a voltage of several hundred volts.

(24) The wire harness **10** includes a wire harness main body **11**. The wire harness main body **11** includes an electric wire member **20** (electric wire) and a tubular exterior member **30** (exterior tube) that encloses the outer circumferential surface of the electric wire member **20**. The wire harness **10** has connectors **C1** and **C2** that are respectively attached to two end portions of the electric wire member **20**. One end portion in the lengthwise direction of the electric wire member **20** is connected to the inverter **M1** via the connector **C1**, and the other end portion in the lengthwise direction of the electric wire member **20** is connected to the high-voltage battery **M2** via the connector **C2**.

(25) As shown in FIGS. **2** and **3**, the wire harness **10** includes a first path restricting member **40** (first path restrictor) that is attached to the outer circumferential surface of the exterior member **30** and a second path restricting member **50** (second path restrictor) that is attached to the outer circumferential surface of the exterior member **30**. The first path restricting member **40** and the second path restricting member **50** restrict the path along which the wire harness main body **11** is routed. Note that the first path restricting member **40** and the second path restricting member **50** are omitted from FIG. **1**.

(26) Configuration of Electric Wire Member 20

(27) As shown in FIGS. **3** and **4**, for example, the electric wire member **20** includes one or more

electric wires **21** (two in the present embodiment) and a braided member **25** that collectively encloses the outer circumferential surfaces of the plurality of electric wires **21**.

(28) As shown in FIG. 4, each electric wire **21** is a coated electric wire that includes a conductive core wire **22** and an insulating coating **23** that encloses the outer circumferential surface of the core wire **22** and has insulating properties. Each electric wire **21** is, for example, a high-voltage electric wire that withstands a high voltage and a large current. Each electric wire **21** may be, for example, a non-shielded electric wire that does not have an electromagnetic shield structure therewith, or a shielded electric wire that has an electromagnetic shield structure therewith. Each electric wire **21** in the present embodiment is a non-shielded electric wire.

(29) As the core wire **22**, a stranded wire that is constituted by a plurality of metal strands twisted together, a single core wire that is constituted by a single conductor, or the like may be used, for example. As the single core wire, a columnar conductor that is constituted by one columnar metal rod with a solid internal structure, a tubular conductor with a hollow internal structure, or the like may be used, for example. As the core wire **22**, a combination of a stranded wire, a columnar conductor, or a tubular conductor may be used. As the material of the core wire **22**, a metal material such as a copper-based material or an aluminum-based material may be used, for example.

(30) The insulating coating **23** covers, for example, all the way around the outer circumferential surface of the core wire **22**. The insulating coating **23** is constituted by, for example, a resin material that has insulating properties.

(31) The cross-sectional shape of each wire **21** cut along a plane that is orthogonal to the lengthwise direction of each wire **21**, that is, the transverse cross-sectional shape of each wire **21**, may be any shape. The transverse cross-sectional shape of each electric wire **21** may be, for example, a circular shape, a semicircular shape, a polygonal shape, a square shape, a flat shape, or the like. The transverse cross-sectional shape of each electric wire **21** in the present embodiment is a circular shape.

(32) The braided member **25** has, for example, a tubular shape that collectively encloses the outer circumferential surfaces of the plurality of electric wires **21**. As the braided member **25**, a braided wire in which a plurality of metal strands are braided or a braided wire in which metal strands and resin strands are braided in combination with each other may be used, for example. As the material of the metal strands, a metal material such as a copper-based material or an aluminum-based material may be used, for example. Although not shown in the drawings, the two end portions of the braided member **25** in the lengthwise direction are grounded at, for example, the connectors **C1** and **C2** (see FIG. 1).

(33) Configuration of Exterior Member **30**

(34) As shown in FIG. 3, the exterior member **30** has a tubular shape that encloses all the way around the outer circumferential surface of the electric wire member **20**. The exterior member **30** in the present embodiment is formed in a cylindrical shape. The exterior member **30** is, for example, provided with a circumferential wall that is formed so as to be continuous all the way around the circumferential surface of the exterior member **30**. The exterior member **30** is, for example, sealed all the way around the circumferential surface of the exterior member **30**. The exterior member **30** has, for example, a function of protecting the electric wire member **20** from flying objects and water droplets.

(35) The exterior member **30** has, for example, flexibility and is easily bendable. Examples of the flexible exterior member **30** include a resin corrugated tube and a rubber waterproof cover. The exterior member **30** of the present embodiment is a resin corrugated tube having an accordion structure in which ring-shaped protrusions **31** and ring-shaped recesses **32** are provided alternately and continuously along the lengthwise direction of the exterior member **30**. Each of the ring-shaped protrusions **31** and the ring-shaped recesses **32** has, for example, a ring shape that encircles the exterior member **30** in the circumferential direction. As the material of the exterior member **30**, for example, a synthetic resin such as polyolefin, polyamide, polyester, or ABS resin

can be used. Note that in FIGS. 1 and 2, the exterior member 30 is shown simplified.

(36) Configuration of First Path Restricting Member 40 and Second Path Restricting Member 50

(37) As shown in FIG. 2, each of the first path restricting member 40 and the second path restricting member 50 holds the exterior member 30. Each of the first path restricting member 40 and the second path restricting member 50 is harder than, for example, the exterior member 30. Each of the first path restricting member 40 and the second path restricting member 50 has a hardness that makes it more difficult to bend in a direction orthogonal to the lengthwise direction of the wire harness main body 11 compared to the exterior member 30. As a result, each of the first path restricting member 40 and the second path restricting member 50 restricts the path of the wire harness main body 11. For example, each of the first path restricting member 40 and the second path restricting member 50 assists the exterior member 30 so that the wire harness main body 11 does not deviate from the desired path by bending due to its own weight or the like. The exterior member 30 is more difficult to bend than, for example, in a state where the first path restricting member 40 and the second path restricting member 50 are not attached.

(38) The first path restricting member 40 is partially provided along the lengthwise direction of the wire harness main body 11. For example, the first path restricting member 40 is attached to the outer circumferential surface of the exterior member 30 along a straight section 11A, which is a straight section of the path of the wire harness main body 11. The first path restricting member 40 restricts the path of the wire harness main body 11 in the straight section 11A. Here, the straight section 11A is a section in which the path of the wire harness main body 11 extends straight in one direction. Note that one or more first path restricting members 40 are provided depending on the path of the wire harness main body 11.

(39) The second path restricting member 50 is provided along a portion in the lengthwise direction of the wire harness main body 11. For example, the second path restricting member 50 is attached to the outer circumferential surface of the exterior member 30 along a bent section 11B, which is a bent section of the path of the wire harness main body 11. The second path restricting member 50 restricts the path of the wire harness main body 11 in the bent section 11B. Here, the bent section 11B is a section in which the path of the wire harness main body 11 is bent two-dimensionally or three-dimensionally. Note that one or more second path restricting members 50 may be provided depending on the path of the wire harness main body 11.

(40) Configuration of First Path Restricting Member 40

(41) As shown in FIG. 4, the first path restricting member 40 covers part of the outer circumferential surface of the exterior member 30 in the circumferential direction of the exterior member 30. The first path restricting member 40 has a tubular shape that covers the outer circumferential surface of the exterior member 30 in a portion in the circumferential direction of the exterior member 30. The transverse cross-sectional shape of the first path restricting member 40 is C-shaped overall. The first path restricting member 40 covers, for example, a range larger than half of the outer circumferential surface of the exterior member 30. The transverse cross-sectional shape of the first path restricting member 40 is, for example, uniform over the entire length in the lengthwise direction of the first path restricting member 40. As shown in FIG. 2, the first path restricting member 40 extends along the path of the straight section 11A, and is formed in a shape extending linearly in one direction, for example.

(42) The first path restricting member 40 is made of metal or resin, for example. The first path restricting member 40 of this embodiment is made of resin. Synthetic resins such as polypropylene, polyamide, and polyacetal can be used as the material of the first path restricting member 40, for example. The first path restricting member 40 can be manufactured through a known manufacturing method such as extrusion molding or injection molding.

(43) As shown in FIG. 3, the first path restricting member 40 has a first main body portion 41 (first main body) that covers part of the outer circumferential surface of the exterior member 30, and an insertion port 42 that opens in a direction orthogonal to the lengthwise direction of the first main

body portion **41**. The first path restricting member **40** has a first end portion **43** and a second end portion **44** which are both ends in the circumferential direction of the first main body portion **41** and form an insertion port **42**.

(44) As shown in FIG. **4**, the first main body portion **41** forms the main part of the first path restricting member **40**. The thickness in the radial direction of the first main body portion **41** is, for example, uniform in the circumferential direction of the first path restricting member **40**. The transverse cross-sectional shape of the first main body portion **41** is formed, for example, in a shape that extends along the outer surface of the exterior member **30**. The transverse cross-sectional shape of the first main body portion **41** is formed, for example, in an arc shape.

(45) The first end portion **43** and the second end portion **44** are provided on mutually opposite sides in the circumferential direction of the first main body portion **41**. The first end portion **43** and the second end portion **44** are provided spaced apart from each other with the insertion port **42** interposed therebetween in the circumferential direction of the first main body portion **41**. In other words, the gap between the first end portion **43** and the second end portion **44** in the circumferential direction of the first path restricting member **40** is formed as the insertion port **42**. As described above, the first path restricting member **40** is formed in a C shape having the insertion port **42** in a portion in the circumferential direction of the first main body portion **41**.

(46) The opening width of the insertion port **42**, that is, the shortest distance between the first end portion **43** and the second end portion **44** is smaller than, for example, the outer diameter of the exterior member **30**. As shown in FIG. **3**, the insertion port **42** extends along the lengthwise direction of the first main body portion **41** over the entire length in the lengthwise direction of the first main body portion **41**. That is, the insertion port **42** is formed so as to be open in a direction orthogonal to the lengthwise direction of the first main body portion **41** and to be open at both ends in the lengthwise direction of the first main body portion **41**.

(47) By inserting the exterior member **30** into the insertion port **42** from a direction orthogonal to the lengthwise direction of the first path restricting member **40**, the first path restricting member **40** is elastically deformed and the opening width of the insertion port **42** is increased. When the exterior member **30** is inserted into the first path restricting member **40**, the first path restricting member **40** elastically returns to its original shape. As a result, the opening width of the insertion port **42** is smaller than the outer diameter of the exterior member **30**, and therefore the first path restricting member **40** is attached to the outer circumferential surface of the exterior member **30**.

(48) As shown in FIG. **4**, the first end portion **43** has a leading end **43A**. The second end portion **44** has a leading end **44A**. The leading ends **43A** and **44A** form an insertion port **42**. The leading ends **43A** and **44A** are curved in a view from the lengthwise direction of the first path restricting member **40**. That is, the transverse cross-sectional shapes of the leading ends **43A** and **44A** are curved.

(49) The first path restricting member **40** has, for example, protruding portions **45** that protrude from the inner surfaces of the first end portion **43** and the second end portion **44**, respectively. Each protruding portion **45** protrudes toward the exterior member **30** inserted in the first path restricting member **40** and can come into contact with the outer surface of the exterior member **30**. Each protruding portion **45** is in contact with the outer surface of the ring-shaped protrusion **31** of the exterior member **30**, for example. The two protruding portions **45**, for example, protrude from the inner surfaces of the leading ends **43A** and **44A**. The transverse cross-sectional shape of each projecting portion **45** is, for example, curved. The cross-sectional shape of each protruding portion **45** of this embodiment is formed in a semicircular shape. Each protruding portion **45** extends over the entire length in the lengthwise direction of the first path restricting member **40** along the lengthwise direction of the first path restricting member **40**, for example.

(50) Each protruding portion **45** presses the exterior member **30** from the outside of the exterior member **30**, for example. The exterior member **30** is elastically sandwiched by the two protruding portions **45** and the first main body portion **41**. As a result, the connection of the first path restricting member **40** to the exterior member **30** is strengthened. Accordingly, the first path

restricting member **40** attached to the outer circumferential surface of the exterior member **30** is suppressed from moving in the lengthwise direction of the exterior member **30**.

(51) As shown in FIG. 3, the first path restricting member **40** has a connection portion **47** (connection) that is connected to the second path restricting member **50**. The connection portion **47** is provided, for example, at an end portion in the lengthwise direction of the first path restricting member **40**. The connection portion **47** has a first end surface **47A** in the lengthwise direction of the first path restricting member **40**.

(52) Configuration of Second Path Restricting Member **50**

(53) As shown in FIG. 2, the second path restricting member **50** is attached to the outer circumferential surface of the exterior member **30** at the bent section **11B**. The second path restricting member **50** is bent along the shape of the bent section **11B**. The second path restricting member **50** is made of metal or resin, for example. The second path restricting member **50** of this embodiment is made of resin. As the material of the second path restricting member **50**, for example, synthetic resin such as polypropylene, polyamide, and polyacetal can be used. The second path restricting member **50** can be manufactured, for example, through a known manufacturing method such as injection molding.

(54) The second path restricting member **50** has a covering portion **51** (cover) that covers a portion in the lengthwise direction of the first path restricting member **40**, and a path restricting portion **70** that is attached to the outer circumferential surface of the exterior member **30**. The second path restricting member **50** is, for example, a single component in which the covering portion **51** and the path restricting portion **70** are formed in one piece. The covering portion **51** is provided, for example, at one end portion in the lengthwise direction of the second path restricting member **50**. The covering portion **51** has a second main body portion **52** (second main body) and a lid portion **56** (lid) coupled to the second main body portion **52**. The path restricting portion **70** has a third main body portion **71** (third main body). The third main body portion **71** is formed in one piece continuously with the second main body portion **52**.

(55) Configuration of Covering Portion **51**

(56) As shown in FIG. 2, the covering portion **51** is provided, for example, so as to overlap with the connecting portion **47** in the radial direction of the exterior member **30**. That is, in the wire harness **10**, one end portion in the lengthwise direction of the first path restricting member **40** and one end portion in the lengthwise direction of the second path restricting member **50** are provided so as to overlap with each other in the lengthwise direction of the wire harness main body **11**.

(57) The covering portion **51** encloses, for example, the outer circumferential surface of the connecting portion **47** of the first path restricting member **40**. The covering portion **51** is formed, for example, in a ring shape that encloses, in the circumferential direction, the entirety of the outer circumferential surface of the connecting portion **47** and the entirety of the outer circumferential surface of the exterior member **30** at the portion to which the connecting portion **47** is attached. The covering portion **51** extends, for example, along the path of the straight section **11A**, and is formed in a shape extending linearly in one direction.

(58) As shown in FIG. 4, the second main body portion **52** covers, for example, a portion of the outer circumferential surface of the exterior member **30** in the circumferential direction of the exterior member **30**. The second main body portion **52** covers, for example, a portion of the outer circumferential surface of the first path restricting member **40** in the circumferential direction of the first path restricting member **40**. The second main body portion **52** has a tubular shape that covers the outer circumferential surface of the first path restricting member **40** at a portion in the circumferential direction of the first path restricting member **40**. The second main body portion **52** covers the outer circumferential surface of the first main body portion **41** of the first path restricting member **40**, for example. The second main body portion **52** covers, for example, a range larger than half of the outer circumferential surface of the first main body portion **41**. The transverse cross-sectional shape of the inner surface of the second main body portion **52** is formed in a shape

corresponding to the outer surface of the first main body portion **41**, for example. The transverse cross-sectional shape of the inner surface of the second main body portion **52** is, for example, curved. The transverse cross-sectional shape of the inner surface of the second main body portion **52** is U-shaped overall.

(59) As shown in FIG. 3, the second main body portion **52** has a second end surface **51A** in the lengthwise direction of the covering portion **51**. The second end surface **51A** is an end surface in the lengthwise direction of the second path restricting member **50** and is an end surface of the covering portion **51**. The second end surface **51A** is provided at the end portion on the side opposite to the end portion connected to the path restricting portion **70** out of the two end portions in the lengthwise direction of the covering portion **51**. The second main body portion **52** has a third end surface **53** and a fourth end surface **54**, which are the two end surfaces in the circumferential direction of the second main body portion **52**. The second main body portion **52** has a first accommodation port **55** formed by the third end surface **53** and the fourth end surface **54**. The first accommodation port **55** is an opening between the third end surface **53** and the fourth end surface **54**. The first accommodation port **55** is open in a direction orthogonal to the lengthwise direction of the second main body portion **52**. The first accommodation port **55** extends over the entire length in the lengthwise direction of the second main body portion **52**, along the lengthwise direction of the second main body portion **52**. The opening width of the first accommodation port **55**, that is, the shortest distance between the third end surface **53** and the fourth end surface **54**, is, for example, equal to the outer diameter of the first path restricting member **40** or larger than the outer diameter of the first path restricting member **40**. The exterior member **30** and the first path restricting member **40** are inserted into the first accommodation port **55** along a direction orthogonal to the lengthwise direction of the second path restricting member **50**.

(60) The lid portion **56** is formed in one piece with the second main body portion **52**, for example. The transverse cross-sectional shape of the inner surface of the lid portion **56** is formed in a shape corresponding to the outer surface of the exterior member **30**, for example. The transverse cross-sectional shape of the inner surface of the lid portion **56** is formed in a shape corresponding to the outer surface of the first path restricting member **40**, for example. The transverse cross-sectional shape of the inner surface of the lid portion **56** is, for example, an elliptical arc or an elongated circular arc. The lid portion **56** is formed in, for example, a substantially semi-elliptical cylindrical shape. Here, an “elongated circle” in this specification is a shape composed of two parallel lines of equal length and two semicircles.

(61) The lid portion **56** covers the first accommodation port **55** of the second main body portion **52**. The lid portion **56** covers the first accommodation port **55** in only the covering portion **51** in the lengthwise direction of the second path restricting member **50**, for example. In other words, the lid portion **56** is provided on only the covering portion **51** of the second path restricting member **50**, and is not provided on the path restricting portion **70**.

(62) The covering portion **51** has, for example, a hinge portion **57** that connects the second main body portion **52** and the lid portion **56**. The hinge portion **57** connects one end in the circumferential direction of the second main body portion **52** and one end in the circumferential direction of the lid portion **56**. One or more (two in this embodiment) lock portions **58** are provided at the other end in the circumferential direction of the second main body portion **52**. One or more (two in this embodiment) claw portions **59** are provided at the other end in the circumferential direction of the lid portion **56**.

(63) The second main body portion **52** and the lid portion **56** are pivotable relative to each other about the hinge portion **57**. The lid portion **56** is pivotable about the hinge portion **57** between an open position shown in FIG. 3 and a closed position shown in FIG. 2. As shown in FIG. 4, the claw portion **59** catches on the lock portion **58** when the lid portion **56** is in the closed position. As a result, the lid portion **56** is held in the closed position. In this manner, the second main body portion **52** and the lid portion **56** are coupled to each other. In a state in which the second main body

portion **52** and the lid portion **56** are coupled to each other, the covering portion **51** has a ring shape that surrounds the exterior member **30** and the outer circumferential surface of the connection portion **47** together. The lid portion **56** covers the first accommodation port **55** of the second main body portion **52** in the closed position. For example, the lid portion **56** covers the insertion port **42** at the connection portion **47** in the closed position.

(64) As shown in FIG. 3, the covering portion **51** has, for example, a restricting wall **60** that can engage with the first end surface **47A** in the lengthwise direction of the first path restriction member **40** in the lengthwise direction of the wire harness main body **11**. The restricting wall **60** protrudes inward in the radial direction of the covering portion **51** from the inner surface of the covering portion **51**, for example. The restricting wall **60** extends along the circumferential direction of the covering portion **51**, for example. The restricting wall **60** extends, for example, over the entire circumferential surface in the circumferential direction of the covering portion **51**. The restricting wall **60** is provided at one end portion in the lengthwise direction of the covering portion **51**.

(65) The restricting wall **60** has, for example, a body-side restricting wall **61** formed on the inner surface of the second main body portion **52** and a lid-side restricting wall **62** formed on the inner surface of the lid portion **56**. The body-side restricting wall **61** and the lid-side restricting wall **62** are provided at the same positions as each other in the lengthwise direction of the wire harness main body **11**, for example.

(66) As shown in FIG. 5, the body-side restricting wall **61** protrudes inward in the radial direction from the inner surface of the second main body portion **52**. The body-side restricting wall **61** extends over the entire circumferential surface of the second main body portion **52** in the circumferential direction of the covering portion **51**, for example. The transverse cross-sectional shape of the inner surface of the body-side restricting wall **61** is formed in the same shape as the transverse cross-sectional shape of the inner surface of the second main body portion **52**. The transverse cross-sectional shape of the inner surface of the body-side restricting wall **61** of this embodiment is formed in a U shape. The transverse cross-sectional shape of the inner surface of the body-side restricting wall **61** is, for example, one size smaller than the transverse cross-sectional shape of the inner surface of the second main body portion **52**.

(67) The body-side restricting wall **61** has a second accommodation port **63** that is open in a direction orthogonal to the lengthwise direction of the second main body portion **52**. The second accommodation port **63** is in communication with the first accommodation port **55**. The opening width of the second accommodation port **63** is smaller than the opening width of the first accommodation port **55**. The opening width of the second accommodation port **63** is smaller than the outer diameter of the first path restricting member **40**, for example.

(68) As shown in FIG. 3, the body-side restricting wall **61** is provided, for example, at the end portion connected to the third main body portion **71** among the two end portions in the lengthwise direction of the second main body portion **52**. The body-side restricting wall **61** extends along the lengthwise direction of the wire harness main body **11**, for example. The body-side restricting wall **61** extends, for example, from the end portion of the second main body portion **52** over the entire length in the lengthwise direction of the third main body portion **71**. For example, at the boundary between the inner surface of the second main body portion **52** and the inner surface of the third main body portion **71**, a level difference is formed by the body-side restricting wall **61**.

(69) As shown in FIG. 6, the end surface **61A** in the lengthwise direction of the body-side restricting wall **61** opposes the first main body portion **41** of the first path restricting member **40** in the lengthwise direction of the wire harness main body **11**, for example, and can come into contact with the first main body portion **41**. The end surface **61A** of the body-side restricting wall **61** opposes the first end surface **47A** in the lengthwise direction of the first path regulation member **40** in the lengthwise direction of the wire harness main body **11**, for example, and can come into contact with the first end surface **47A**.

(70) As shown in FIG. 5, the lid-side restricting wall **62** protrudes inward in the radial direction from the inner surface of the lid portion **56**. The lid-side restricting wall **62** extends over the entire circumferential surface of the lid portion **56** in the circumferential direction of the covering portion **51**, for example. The transverse cross-sectional shape of the inner surface of the lid-side restricting wall **62** is formed in a shape corresponding to the outer surface of the first path restricting member **40**, for example. The transverse cross-sectional shape of the inner surface of the lid-side restricting wall **62** is formed, for example, in an arc shape or an elliptical arc shape. The transverse cross-sectional shape of the inner surface of the lid-side restricting wall **62** of this embodiment is formed in a semi-elliptical arc shape.

(71) The lid-side restricting wall **62** opposes the first end portion **43** and the second end portion **44** of the first path restricting member **40** in the lengthwise direction of the wire harness main body **11** (see FIG. 3), for example, and can come into contact with the first end portion **43** and the second end portion **44**.

(72) As shown in FIG. 3, the restricting wall **60** composed of the body-side restricting wall **61** and the lid-side restricting wall **62** restricts movement of the first path restricting member **40** in the lengthwise direction of the wire harness main body **11** by coming into contact with the first end surface **47A** in the lengthwise direction of the first path restricting member **40**. For example, the restricting wall **60** restricts movement of the first path restricting member **40** so that the first path restricting member **40** is not inserted into the path restricting portion **70**. Such a restricting wall **60** functions as a second mark indicating a maximum insertion amount of the first path restriction member **40** into the covering portion **51** in the lengthwise direction of the wire harness main body **11**. That is, the restricting wall **60** functions as a mark indicating the maximum insertion amount by which the first path restricting member **40** can be allowed to be inserted into the covering portion **51**, that is, the maximum overlapping width between the connection portion **47** and the covering portion **51** in the lengthwise direction of the wire harness main body **11**. In other words, the restricting wall **60** functions as a mark indicating the limit position to which the first path restricting member **40** can be allowed to be inserted into the covering portion **51** in the lengthwise direction of the wire harness main body **11**.

(73) As shown in FIG. 4, the covering portion **51** has, for example, a projecting portion **65**. The projecting portion **65** protrudes from the inner surface of the covering portion **51** toward the insertion port **42** of the first path restricting member **40** inserted in the covering portion **51**. The projecting portion **65** protrudes so as to be arranged inside the insertion port **42**. The projecting portion **65** can come into contact with the first end portion **43** and the second end portion **44** in the circumferential direction of the first path restricting member **40**. Due to the projecting portion **65** and the first end portion **43** or the second end portion **44** being engaged with each other, relative movement of the first path restricting member **40** with respect to the covering portion **51** in the circumferential direction of the first path restricting member **40** is suppressed. For this reason, it is possible to suppress misalignment of the first path restricting member **40** with respect to the covering portion **51** in the circumferential direction of the first path restricting member **40**.

(74) The projecting portion **65** is provided on the inner surface of the lid portion **56**, for example. The projecting portion **65** protrudes from the inner surface of the lid portion **56** toward the first accommodation port **55**, for example. For this reason, when the projecting portion **65** is arranged inside the insertion port **42**, the insertion port **42** faces the lid portion **56** side of the covering portion **51**.

(75) As shown in FIG. 3, the projecting portion **65** extends along the lengthwise direction of the wire harness main body **11**, for example. The projecting portion **65** extends over the entire length in the lengthwise direction of the covering portion **51**, along the lengthwise direction of the covering portion **51**, for example. The projecting portion **65** is partially provided in the circumferential direction of the lid portion **56**, for example. The projecting portion **65** is provided, for example, at an intermediate portion in the circumferential direction of the lid portion **56**. A portion of the

projecting portion **65** constitutes, for example, the lid-side restricting wall **62**. For example, the projecting portion **65** provided at one end portion in the lengthwise direction of the covering portion **51** constitutes the lid-side restricting wall **62**.

(76) As shown in FIG. **6**, the projecting portion **65** has, for example, a protrusion **66** that fits into a ring-shaped recess **32** of the exterior member **30**. The protrusion **66** is provided so as to further protrude from the projecting leading end surface of the projecting portion **65**. A plurality of protrusions **66** are provided in the lengthwise direction of the wire harness main body **11**. In this embodiment, four protrusions **66** are provided in the lengthwise direction of the wire harness main body **11**, and each of the protrusions **66** is inserted into a different ring-shaped recess **32**. Due to the protrusions **66** entering the ring-shaped recesses **32**, movement of the exterior member **30** relative to the second path restricting member **50** in the lengthwise direction of the exterior member **30** can be suppressed.

(77) As shown in FIG. **3**, the covering portion **51** has the mark **67** indicating the necessary minimum insertion amount of the first path restricting member **40** into the covering portion **51** in the lengthwise direction of the wire harness main body **11**. The mark **67** is provided on the inner surface of the second main body portion **52**, for example.

(78) As shown in FIG. **6**, the mark **67** has, for example, a recess **68** provided on the inner surface of the second main body portion **52**. The recess **68** is formed so as to be recessed from the inner surface of the second main body portion **52** toward the outside in the radial direction of the covering portion **51**. The recess **68** is, for example, a recess that does not pass through the second main body portion **52** in the radial direction of the covering portion **51**. The recess **68** is partially provided in the lengthwise direction of the covering portion **51**. The recess **68** is provided, for example, in the intermediate portion in the lengthwise direction of the covering portion **51**. The recess **68** is provided spaced apart by, for example, a first distance **L1** from the second end surface **51A** in the lengthwise direction of the covering portion **51** in the lengthwise direction of the wire harness main body **11**. Here, the second end surface **51A** is provided at the end portion on the side opposite to the restricting wall **60** out of the two end portions in the lengthwise direction of the covering portion **51**. Also, the first distance **L1** is set, for example, to a distance corresponding to the needed minimum insertion amount of the first path restricting member **40** into the covering portion **51** in the lengthwise direction of the wire harness main body **11**. The minimum insertion amount of the first path restricting member **40** is, for example, an insertion amount that is needed to make it difficult for the first path restricting member **40** to detach from the covering portion **51**, that is, the overlapping width between the connection portion **47** and the covering portion **51** in the lengthwise direction of the wire harness main body **11**. The minimum insertion amount of the first path restricting member **40** is set, for example, according to the ease of movement of the first path restricting member **40** with respect to the covering portion **51** in the lengthwise direction of the wire harness main body **11**. For example, if the first path restricting member **40** is freely movable with respect to the covering portion **51** in the lengthwise direction of the wire harness main body **11**, the minimum insertion amount of the first path restricting member **40** is set to be large. The recess **68** of the present embodiment is provided at a position closer to the restricting wall **60** out of the second end surface **51A** and the restricting wall **60**, in the lengthwise direction of the covering portion **51**. That is, the recess **68** of the present embodiment is provided so as to be biased toward the restricting wall **60** in the lengthwise direction of the covering portion **51**. The recess **68** of this embodiment is provided spaced apart from the restricting wall **60** in the lengthwise direction of the covering portion **51**.

(79) As shown in FIGS. **3** and **4**, the recess **68** extends along the circumferential direction of the covering portion **51**, for example. The recess **68** extends over the entire circumferential surface of the second main body portion **52** in the circumferential direction of the covering portion **51**, for example.

(80) Structure of Path Restricting Portion **70**

(81) As shown in FIG. 2, the third main body portion **71** of the path restricting portion **70** extends, for example, along the path of the bent section **11B**. That is, the third main body portion **71** has a bent shape extending along the shape of the bent section **11B**. The third main body portion **71** covers, for example, a portion of the outer circumferential surface of the exterior member **30** in the circumferential direction of the exterior member **30**. The third main body portion **71** has a tubular shape that covers the outer circumferential surface of the exterior member **30** at a portion in the circumferential direction of the exterior member **30**. The third main body portion **71** covers, for example, a range larger than half of the outer circumferential surface of the exterior member **30**. The transverse cross-sectional shape of the inner surface of the third main body portion **71** is formed in a shape corresponding to the outer surface of the exterior member **30**. The transverse cross-sectional shape of the inner surface of the third main body portion **71** is, for example, curved. The transverse cross-sectional shape of the inner surface of the third main body portion **71** is U-shaped overall.

(82) The third main body portion **71** has a second accommodation port **63** that is open in a direction orthogonal to the lengthwise direction of the third main body portion **71**. The second accommodation port **63** extends over the entire length in the lengthwise direction of the third main body portion **71**, along the lengthwise direction of the third main body portion **71**. The second accommodation port **63** is in communication with the first accommodation port **55**. The accommodation port constituted by the first accommodation port **55** and the second accommodation port **63** extends over the entire length in the lengthwise direction of the second path restricting member **50**, along the lengthwise direction of the second path restricting member **50**. That is, the accommodation port constituted by the first accommodation port **55** and the second accommodation port **63** is open in a direction orthogonal to the lengthwise direction of the second path restricting member **50** and is open at both ends in the lengthwise direction of the second path restricting member **50**.

(83) The opening width of the second accommodation port **63** is equal to the outer diameter of the exterior member **30** or larger than the outer diameter of the exterior member **30**, for example. The exterior member **30** is inserted into the second accommodation port **63** along a direction orthogonal to the lengthwise direction of the second path restricting member **50**. Note that the first path restricting member **40** is not inserted in the third main body portion **71**.

(84) Configuration of Fixing Members **81** and **82**

(85) The wire harness **10** has, for example, a fixing member **81** that fixes the first path restricting member **40** to the outer circumferential surface of the exterior member **30**. The wire harness **10** has, for example, a fixing member **82** that fixes the second path restricting member **50** to the outer circumferential surface of the exterior member **30**. As the fixing members **81** and **82**, for example, a cable tie made of resin or metal, a crimping ring, adhesive tape, or the like can be used. The fixing members **81** and **82** of this embodiment are adhesive tape.

(86) The fixing member **81** is formed, for example, so as to fix the end portion provided on the side opposite to the connecting portion **47** out of the end portions in the lengthwise direction of the first path restricting member **40** to the outer surface of the exterior member **30**. The fixing member **81** is wound, for example, from the outer surface of the end portion in the lengthwise direction of the first path restricting member **40** to the outer surface of the exterior member **30**. Accordingly, it is possible to suppress movement of the first path restricting member **40** with respect to the exterior member **30** in the lengthwise direction and the circumferential direction of the wire harness body **11**. The fixing member **82** is formed, for example, so as to fix the end portion provided on the side opposite to the covering portion **51** among the end portions in the lengthwise direction of the second path restricting member **50** to the outer surface of the exterior member **30**. The fixing member **82** is wound, for example, from the outer surface of the end portion in the lengthwise direction of the second path restricting member **50** to the outer surface of the exterior member **30**. This makes it possible to suppress movement of the second path restricting member **50** with respect

to the exterior member **30** in the lengthwise direction and the circumferential direction of the wire harness body **11**.

(87) Next, actions of this embodiment will be described.

(88) The connection portion **47** of the first path restricting member **40** and the covering portion **51** of the second path restricting member **50** are provided so as to overlap each other in the lengthwise direction of the wire harness main body **11**. At this time, the covering portion **51** covering the outer circumferential surface of the connection portion **47** is provided with a mark **67** indicating the necessary minimum insertion amount of the first path restricting member **40** into the covering portion **51**. According to this configuration, during the task of inserting the first path restricting member **40** into the covering portion **51**, the position corresponding to the minimum insertion amount of the first path restricting member **40** in the covering portion **51** can be easily confirmed based on the mark **67**. That is, based on the mark **67**, it is possible to easily confirm how far the first path restricting member **40** needs to be inserted into the covering portion **51**. Here, when inserting the first path restricting member **40** into the covering portion **51**, for example, the first path restricting member **40** is inserted into the covering portion **51** such that the first end surface **47A** of the first path restricting member **40** exceeds the mark **67** while the mark **67** is visually confirmed. That is, while the mark **67** is visually confirmed, the first path restricting member **40** is inserted into the covering portion **51** such that the first end surface **47A** is inserted farther than the mark **67** in the lengthwise direction of the wire harness main body **11**, that is, to a position close to the restricting wall **60** relative to the mark **67**. In other words, the first path restricting member **40** is inserted into the covering portion **51** such that the first path restricting member **40** hides the recess **68** that is the mark **67**, that is, such that the recess **68** cannot be seen. By inserting the first path restricting member **40** into the covering portion **51** farther than the mark **67** in this manner, the first path restricting member **40** can be suitably inserted into the covering portion **51** by a necessary minimum insertion amount or more. As a result, detachment of the first path restricting member **40** from the covering portion **51** can be suitably suppressed.

(89) Next, the effects of this embodiment will be described.

(90) (1) The covering portion **51** covering the outer circumferential surface of the connection portion **47** is provided with the mark **67** indicating the necessary minimum insertion amount of the first path restricting member **40** into the covering portion **51**. According to this configuration, during the task of inserting the first path restricting member **40** into the covering portion **51**, the position corresponding to the minimum insertion amount of the first path restricting member **40** in the covering portion **51** can be easily confirmed based on the mark **67**. For this reason, by inserting the first path restricting member **40** into the covering portion **51** while confirming the mark **67**, the first path restricting member **40** can be suitably inserted into the covering portion **51** by a necessary minimum insertion amount or more. As a result, detachment of the first path restricting member **40** from the covering portion **51** can be suitably suppressed. As a result, the connection reliability between the first path restricting member **40** and the second path restricting member **50** can be improved.

(91) (2) During the task of inserting the first path restricting member **40** into the covering portion **51**, the first path restricting member **40** can be inserted into the covering portion **51** using the mark **67** as a guide. As a result, the efficiency of the task of inserting the first path restricting member **40** into the covering portion **51** can be improved, and therefore the assembly workability of the wire harness **10** can be improved.

(92) (3) In this embodiment, the first path restricting member **40** is inserted into the covering portion **51** such that the mark **67** is hidden. For this reason, by confirming whether or not the mark **67** is visible after inserting the first path restricting member **40**, it is possible to easily confirm whether or not the first path restricting member **40** has been inserted into the covering portion **51** by the minimum necessary insertion amount or more. That is, it is possible to easily confirm whether or not the first path restricting member **40** has been properly inserted into the covering

portion **51** by using the positional relationship between the first path restricting member **40** and the mark **67** as a guide.

(93) (4) The insertion port **42** of the first path restricting member **40** is open in a direction orthogonal to the lengthwise direction of the first path restricting member **40** and extends over the entire length in the lengthwise direction of the first path restricting member **40**. As a result, it is possible to attach the first path restricting member **40** to the exterior member **30** from the insertion port **42** after performing terminal processing such as attachment of the connectors **C1** and **C2** to the end portions in the lengthwise direction of the electric wire members **20**. In this manner, since the first path restricting member **40** can be retrofitted, the assembly workability of the wire harness **10** can be improved.

(94) (5) The covering portion **51** is provided with a restricting wall **60** that can engage with the first end surface **47A** in the lengthwise of the first path regulation member **40**, in the lengthwise direction of the wire harness main body **11**. When the first end surface **47A** of the first path restricting member **40** is engaged with the restricting wall **60**, relative movement of the first path restricting member **40** with respect to the second path restricting member **50** in the lengthwise direction of the wire harness main body **11** is suppressed. For this reason, when the first end surface **47A** is engaged with the restricting wall **60**, insertion of the first path restricting member **40** into the path restricting portion **70** of the second path restricting member **50** can be suppressed. In other words, the restricting wall **60** can function as a mark indicating the maximum insertion amount of the first path restricting member **40** into the covering portion **51** in the lengthwise direction of the wire harness main body **11**. As a result, during the task of inserting the first path restricting member **40** into the covering portion **51**, the first path restricting member **40** can be inserted into the covering portion **51** using the mark **67** and the restricting wall **60** as a guide. This makes it possible to suitably insert the first path restricting member **40** into the covering portion **51** to a desired position by inserting the first path restricting member **40** into the covering portion **51** such that the first end surface **47A** is positioned between the mark **67** and the restricting wall **60**. As a result, the efficiency of the task of inserting the first path restricting member **40** into the covering portion **51** can be further improved, and therefore the assembly workability of the wire harness **10** can be further improved.

(95) (6) When attaching the first path restricting member **40** to the second path restricting member **50**, the first path restricting member **40** can be easily positioned with respect to the second path restricting member **50** in the lengthwise direction of the wire harness main body **11** by engaging the first end surface **47A** in the lengthwise direction of the first path restricting member **40** with the restricting wall **60**.

(96) (7) The mark **67** is provided spaced apart by the first distance **L1** corresponding to the minimum insertion amount from the second end surface **51A** provided on the side opposite to the restricting wall **60** in the lengthwise direction of the covering portion **51**. According to this configuration, the minimum insertion amount and the maximum insertion amount of the first path restricting member **40** into the covering portion **51** can be easily confirmed based on the mark **67** and the restricting wall **60**. Accordingly, the necessary insertion amount of the first path restricting member **40** into the covering portion **51** can be easily confirmed based on the mark **67** and the restricting wall **60**.

(97) (8) The covering portion **51** is formed in a ring shape that surrounds the connection portion **47** and the exterior member **30** together. The covering portion **51** has a second main body portion **52** that covers a portion of the outer circumferential surface of the first main body portion **41** and a lid portion **56** that is coupled to the second main body portion **52**. According to this configuration, the covering portion **51** is formed in an annular shape surrounding the connection portion **47** and the exterior member **30**, and the covering portion **51** is divided into the second main body portion **52** and the lid portion **56**, and therefore the second path restricting member **50** including the covering portion **51** can be retrofitted to the first path restricting member **40** and the exterior member **30**.

This makes it possible to further improve the assembly workability of the wire harness **10**.

(98) (9) The mark **67** has a recess **68** provided on the inner surface of the second main body portion **52**. According to this configuration, when inserting the first path restricting member **40** into the second main body portion **52**, the recess **68** serving as the mark **67** is provided on the inner surface of the second main body portion **52**, and therefore the position of the recess **68** is easy to visually confirm. Furthermore, the positional relationship between the recess **68** and the first end surface **47A** of the first path restricting member **40** can be easily confirmed visually. As a result, the workability when inserting the first path restricting member **40** into the covering portion **51** can be improved, and the first path restricting member **40** can be suitably inserted into the covering portion **51** by the necessary minimum insertion amount or more.

(99) (10) The recess **68** extends over the entirety of the circumferential surface of the second main body portion **52** in the circumferential direction of the covering portion **51**. According to this configuration, since the recess **68** is formed in a wide range in the circumferential direction of the covering portion **51**, the position of the recess **68** is easier to confirm visually.

(100) (11) The recess **68** is a recess that does not pass through the second main body portion **52** in the radial direction of the covering portion **51**. According to this configuration, the recess **68** does not pass through the second main body portion **52** in the radial direction, and therefore the recess **68** can be formed in a wide range in the circumferential direction of the covering portion **51**. Also, since the recess **68** is provided on the inner surface of the second main body portion **52**, it is possible to suitably suppress a case in which the recess **68** hinders the insertion of the first path restricting member **40** into the second main body portion **52**.

(101) (12) The first path restricting member **40** restricts the path of the straight section **11A**, and the second path restricting member **50** restricts the path of the bent section **11B**. This makes it possible to suppress a case in which the path of the straight section **11A** and the path of the bent section **11B** deviate from desired paths.

Other Embodiments

(102) The above embodiment can be modified and implemented as follows. The above embodiment and the following modifications can be implemented in combination with each other as long as no contradiction arises.

(103) In the above-described embodiment, the recess **68** was formed so as to extend continuously along the circumferential direction of the covering portion **51**, but there is no limitation to this. For example, the recess **68** may be provided so as to intermittently extend along the circumferential direction of the covering portion **51**. That is, the recess **68** may be partially provided at predetermined intervals in the circumferential direction of the covering portion **51**.

(104) In the above-described embodiment, the recess **68** was formed so as to extend over the entire circumferential surface of the second main body portion **52** in the circumferential direction of the covering portion **51**, but there is no limitation to this. For example, the recess **68** may be formed in only a portion of the second main body portion **52** in the circumferential direction of the covering portion **51**.

(105) In the above-described embodiment, the recess **68** was formed as a recess that does not pass through the second main body portion **52** in the radial direction of the covering portion **51**, but there is no limitation to this. For example, the recess **68** may be formed as a through hole passing through the second main body portion **52** in the radial direction.

(106) In the above-described embodiment, the recess **68** was provided on only the inner surface of the second main body portion **52**, but there is no limitation to this. For example, the recess **68** may be provided on the inner surface of the second main body portion **52** and on the inner surface of the lid portion **56** as well.

(107) Although the mark **67** in the above-described embodiment had the recess **68** provided on the inner surface of the second main body portion **52**, there is no limitation to this. For example, instead of the recess **68**, the mark **67** may have a protrusion that protrudes inward in the radial

direction of the covering portion **51** from the inner surface of the second main body portion **52**.

Also, the mark **67** may have a pattern formed by printing or engraving on the inner surface of the second main body portion **52** instead of the recess **68**, for example.

(108) For example, as shown in FIG. 7, the mark **67** may also be provided on the end surface in the circumferential direction of the second main body portion **52**. For example, the mark **67** may also be provided on at least one of the third end surface **53** and the fourth end surface **54** of the second main body portion **52**. The mark **67** of this modified example has a groove portion **91** provided in the third end surface **53**. The groove portion **91** is formed so as to be recessed from the third end surface **53**. The groove portion **91** extends over the entire length of the third end surface **53** in the radial direction of the covering portion **51**, for example.

(109) According to this configuration, since the groove portion **91** serving as the mark **67** is formed in the third end surface **53**, the groove portion **91** can be visually confirmed even after the first path restricting member **40** is inserted into the second main body portion **52**. As a result, even after the first path restricting member **40** is inserted into the second main body portion **52**, the positional relationship between the first end surface **47A** (see FIG. 3) of the first path restricting member **40** and the groove portion **91** can be easily confirmed.

(110) For example, as shown in FIG. 7, the mark **67** may have a protrusion **92** provided on the lid portion **56**. Here, the lid portion **56** has a fifth end surface **56A** and a sixth end surface **56B**, which are both end surfaces in the circumferential direction of the lid portion **56**. The fifth end surface **56A** is provided at the end portion connected to the second main body portion **52** via the hinge portion **57** among the two end portions in the circumferential direction of the lid portion **56**. The protrusion **92** of this modified example is formed so as to protrude from the fifth end surface **56A**. The protrusion **92** is formed, for example, to have a size that allows recess-protrusion fitting in the groove portion **91**. The protrusion **92** is formed so as to fit by recess-protrusion fitting in the groove **91**, for example, when the lid **56** is in the closed position.

(111) In the modified example shown in FIG. 7, the groove portion **91** may be provided in the fourth end surface **54**. Also, the groove portion **91** may be provided on both the third end surface **53** and the fourth end surface **54**.

(112) In the modified example shown in FIG. 7, the protrusion **92** may also be provided on the sixth end surface **56B**. Also, the protrusion **92** may be provided on both the fifth end surface **56A** and the sixth end surface **56B**.

(113) In the modified example shown in FIG. 7, the groove portion **91** was formed in the third end surface **53**, the protrusion **92** was formed in the fifth end surface **56A**, and the groove portion **91** and the protrusion **92** were fitted together through recess-protrusion fitting, but the recess-protrusion relationship may also be reversed. That is, the protrusion **92** may be formed on the third end surface **53** and the groove portion **91** may be formed on the fifth end surface **56A**.

(114) In the modified example shown in FIG. 7, the recess **92** may be changed to a groove portion **91** recessed from the fifth end surface **56A**. That is, the groove portion **91** may be formed in both the third end surface **53** and the fifth end surface **56A**.

(115) In the modified example shown in FIG. 7, the protrusion **92** may be omitted.

(116) The mark **67** of the above-described embodiment may have both the recess **68** shown in FIG. 3 and the groove portion **91** and the protrusion **92** shown in FIG. 7.

(117) In the above-described embodiment, the second mark indicating the maximum insertion amount of the first path restricting member **40** into the covering portion **51** was embodied as the restricting wall **60**, but there is no limitation to this. For example, the second mark is not limited to a structure protruding inward in the radial direction of the covering portion **51** from the inner surface of the second main body portion **52**, but may be changed to a structure recessed outward in the radial direction of the covering portion **51** from the inner surface of the second main body portion **52**. For example, the second mark may be changed to a pattern formed on the inner surface of the second main body portion **52** by printing or engraving. Also, the formation position of the

second mark is not limited to the inner surface of the second main body portion **52**, and may be changed to the third end surface **53** or the fourth end surface **54**, or may be changed to the fifth end surface **56A** or the sixth end surface **56B**.

(118) The first distance **L1** in the above-described embodiment can be changed as appropriate.

(119) For example, as shown in FIG. **8**, the first distance **L1** may be set shorter than the first distance **L1** shown in FIG. **6**. For example, as in the above-described embodiment, when the end portion in the lengthwise direction of the first path restricting member **40** is fixed to the outer surface of the exterior member **30** by the fixing member **81** (see FIG. **2**), movement of the first path restricting member **40** in the lengthwise direction of the wire harness main body **11** is restricted.

For this reason, in this case, the first distance **L1** may be set to be short. When the first distance **L1** is set to be short, the recess **68** that serves as the mark **67** is provided at a position shifted toward the second end surface **51A** in the lengthwise direction of the covering portion **51**, for example.

(120) For example, as shown in FIG. **8**, apart from the mark **67** indicating the minimum insertion amount and the restricting wall **60** indicating the maximum insertion amount, a third mark **93** indicating an insertion amount between the minimum insertion amount and the maximum insertion amount may be provided. The third mark **93** of this modified example is a recess formed so as to be recessed from the inner surface of the second main body portion **52**. The third mark **93** is provided between the mark **67** and the restricting wall **60** in the lengthwise direction of the wire harness main body **11**.

(121) In the modified example shown in FIG. **8**, the structure of the third mark **93** can be modified as appropriate. For example, the third mark **93** is not limited to a structure that is recessed outward in the radial direction of the covering portion **51** from the inner surface of the second main body portion **52**, and may be changed to a structure that protrudes inward in the radial direction of the covering portion **51** from the inner surface of the second main body portion **52**. For example, the third mark **93** may be changed to a pattern formed on the inner surface of the second main body portion **52** by printing or engraving. Also, the formation position of the third mark **93** is not limited to the inner surface of the second main body portion **52**, and may be changed to the third end surface **53** or the fourth end surface **54**, or may be changed to the fifth end surface **56A** or the sixth end surface **56B**.

(122) In the above-described embodiment, second path restricting members **50** may be provided on both sides in the lengthwise direction of the first path restricting member **40**. At this time, for example, the first path restricting member **40** easily moves along the lengthwise direction of the wire harness main body **11** between one second path restricting member **50** and the other second path restricting member **50** in some cases. In this case, for example, the minimum insertion amount of the first path restricting member **40** with respect to the covering portion **51** is set to be large, and the first distance **L1** is set to be long.

(123) The structure of the second path restricting member **50** in the above-described embodiment can be changed as appropriate. For example, other structures are not particularly limited as long as the second path restricting member **50** has a structure that covers the connection portion **47** and has a covering portion **51** provided with the mark **67**.

(124) The protruding portion **66** of the lid portion **56** may be omitted.

(125) The projecting portion **65** of the lid **56** may be omitted.

(126) At least one of the body-side restricting wall **61** and the lid-side restricting wall **62** may be omitted.

(127) In the covering portion **51** of the above-described embodiment, the second main body portion **52** and the lid portion **56** are formed in one piece with each other, but there is no limitation to this, and the second main body portion **52** and the lid portion **56** may also be separate from each other. That is, the second main body portion **52** and the lid portion **56** may be separate components.

(128) The lid portion **56** of the covering portion **51** may also be omitted.

(129) The bent shape of the path restricting portion **70** can be changed as appropriate.

(130) The transverse cross-sectional shapes of the inner surfaces of the second main body portion **52** and the third main body portion **71** are not limited to U shapes, and can be changed to, for example, an arc shape, an elliptical arc shape, or the like.

(131) A lid portion connected to the third main body portion **71** may be provided.

(132) The second path restricting member **50** in the above embodiment was formed so as to restrict the path of the bent section **11B** of the wire harness main body **11**, but there is no limitation to this. For example, the second path restricting member **50** may be changed to a shape that restricts the path of the straight section **11A** of the wire harness main body **11**. The second path restricting member **50** in this case is changed, for example, to a shape in which the curved shape of the path restricting portion **70** extends linearly.

(133) In the above-described embodiment, the attachment member (attachment) having the covering portion **51** was embodied in the second path restricting member **50**, but there is no limitation to this. For example, the attachment member may be also embodied in a structure having only the covering portion **51**.

(134) The structure of the first path restricting member **40** in the above embodiment can be changed as appropriate. For example, as long as the first path restricting member **40** has the insertion port **42** and has a structure that can be attached to the outer circumferential surface of the exterior member **30**, other structures are not particularly limited.

(135) Each protruding portion **45** of the above-described embodiment may be provided at a position farther away from the insertion port **42** than the leading ends **43A** and **44A** in the circumferential direction of the first path restricting member **40**.

(136) Each protruding portion **45** may be partially provided in the lengthwise direction of the first path restricting member **40**.

(137) At least one of the two protruding portions **45** may be omitted.

(138) The first path restricting member **40** may be provided with a second protruding portion that protrudes from the inner surface of the intermediate portion in the circumferential direction of the first main body portion **41** and can come into contact with the outer surface of the exterior member **30**. According to this configuration, the protruding portion **45** and the second protruding portion can be brought into contact with the outer surface of the exterior member **30** together. Therefore, rattling of the first path restricting member **40** can be suppressed.

(139) In the first path restricting member **40**, a groove extending along the lengthwise direction may be provided on the outer surface of the first main body portion **41**. According to this configuration, the first main body portion **41** is easily deformed to the outer peripheral side using the groove as a starting point, whereby the insertion opening **42** can be more easily expanded. As a result, it is possible to contribute to the improvement of the assemblability of the first path restricting member **40**.

(140) The thickness in the radial direction of the first main body portion **41** may be changed in the circumferential direction.

(141) The transverse cross-sectional shape of the first main body portion **41** is not limited to an arc shape, but can be changed to, for example, an elliptical arc shape, a U shape, or the like.

(142) In the above-described embodiment, the first path restricting member **40** and the second path restricting member **50** were harder than the exterior member **30**, but there is no limitation to this, and the hardness may be less than or equal to that of the exterior member **30**. That is, if the first path restricting member **40** and the second path restricting member **50** act so that the wire harness main body **11** is less likely to bend than the wire harness main body **11** in the state where the first path restricting member **40** and the second path restricting member **50** are not attached, the first path restricting member **40** and the second path restricting member **50** need not be harder than the exterior member **30**.

(143) At least one of the restricting members **91** and **92** in the above-described embodiment may be omitted.

- (144) For example, the exterior member **30** in the above-described embodiment may be a resin corrugated tube with a metal layer that contains a metal material, formed on the outer surface thereof.
- (145) The exterior member **30** in the above-described embodiment is not limited to being a corrugated tube, and may be an exterior member that is not provided with an annular protrusion **31** or an annular recess **32**, for example.
- (146) The exterior member **30** in the above-described embodiment may have a slit that extends in the lengthwise direction of the exterior member **30**.
- (147) Although the electric wires **21** in the above-described embodiment are high-voltage electric wires, the present disclosure is not limited to such a configuration. For example, the electric wires **21** may be low-voltage electric wires.
- (148) In the electric wire member **20** in the above-described embodiment, an electromagnetic shield member is embodied as the braided member **25**. However, the present disclosure is not limited to such a configuration. For example, the electromagnetic shield member in the electric wire member **20** may be embodied as a metal foil.
- (149) The braided member **25** of the electric wire member **20** in the above-described embodiment may be omitted.
- (150) In the above-described embodiment, the number of electric wires **21** included in the electric wire member **20** is two. However, the present disclosure is not limited to such a configuration. The number of electric wires **21** may be one or three or more.
- (151) The positional relationship between the inverter **M1** and the high-voltage battery **M2** in the vehicle **V** is not limited to that in the above-described embodiment, and may be changed as appropriate depending on the vehicle configuration.
- (152) In the above-described embodiment, a plurality of on-board devices to which the wire harness **10** is to be electrically connected are embodied as the inverter **M1** and the high-voltage battery **M2**. However, the present disclosure is not limited to such a configuration. The plurality of on-board devices to which the wire harness **10** is to be electrically connected are not particularly limited as long as they are electric devices to be mounted in the vehicle **V**.
- (153) The embodiments disclosed herein are illustrative in all aspects and should not be considered restrictive. The scope of the present disclosure is indicated by the scope of claims, not the above-mentioned meaning, and is intended to include all modifications within the meaning and scope equivalent to the scope of claims.

Claims

1. A wire harness comprising: a wire harness main body including an electric wire and an exterior tube that covers an outer circumferential surface of the electric wire, the exterior tube comprising a plurality of recesses; a path restrictor that is attached to an outer circumferential surface of the exterior tube and restricts a path of the wire harness; and an attachment attached to an outer circumferential surface of a portion in a lengthwise direction of the path restrictor, wherein: the path restrictor includes a first main body that covers a portion of the outer circumferential surface of the exterior tube and an insertion port that is open in a direction orthogonal to a lengthwise direction of the first main body and extends over an entire length in the lengthwise direction of the first main body, the path restrictor has a connection connected to the attachment, the attachment has a cover that covers an outer circumferential surface of the connection, the cover comprises: a mark indicating a necessary minimum insertion amount of the path restrictor into the cover in a lengthwise direction of the wire harness main body; and a plurality of protrusions that fit respectively into the plurality of recesses of the exterior tube, each protrusion being separate from the mark, and the cover has a restricting wall that is engageable with a first end surface of the path restrictor in the lengthwise direction of the path restrictor, in the lengthwise direction of the wire

harness main body.

2. The wire harness according to claim 1, wherein: the restricting wall is provided at one end in a lengthwise direction of the cover, the cover has a second end surface of an end on a side opposite to the restricting wall out of two ends in the lengthwise direction of the cover, the mark is provided spaced apart from the second end surface by a first distance in the lengthwise direction of the wire harness main body, and the first distance is a distance corresponding to the minimum insertion amount.

3. The wire harness according to claim 1, wherein: the cover is formed in a ring shape that encloses, in a circumferential direction, an entirety of the outer circumferential surface of the connection and an entirety of the outer circumferential surface of the exterior tube at a portion where the connection is attached, and the cover has a second main body that covers a portion of the outer circumferential surface of the first main body, and a lid that is coupled to the second main body.

4. The wire harness according to claim 3, wherein the mark is provided on an inner surface of the second main body.

5. The wire harness according to claim 4, wherein the mark extends over an entirety of a circumferential surface of the second main body in a circumferential direction of the cover.

6. The wire harness according to claim 4, wherein: the mark has a recess that is recessed from the inner surface of the second main body, and the recess is a recess that does not pass through the second main body in a radial direction of the cover.

7. The wire harness according to claim 3, wherein: the second main body has a third end surface and a fourth end surface, which are both end surfaces in a circumferential direction of the second main body, and the mark is provided on at least one of the third end surface and the fourth end surface.

8. The wire harness according to claim 1, wherein: the path restrictor is a first path restrictor, the attachment is a second path restrictor that is attached to an outer circumferential surface of the exterior tube and restricts the path of the wire harness main body, the connection is provided at an end in a lengthwise direction of the first path restrictor, the cover is provided at an end in a lengthwise direction of the second path restrictor, the first path restrictor restricts a path of a straight section, which is a portion with a linear shape in the path of the wire harness main body, and the second path restrictor restricts a path of a bend, which is a portion with a bent shape in the path of the wire harness main body.

9. The wire harness according to claim 1, wherein: the minimum insertion amount mark is spaced from an end surface of the cover by a first distance, and each protrusion among the plurality of protrusions is spaced from the end surface of the cover by a distance shorter than the first distance.

10. A wire harness comprising: a wire harness main body including an electric wire and an exterior tube that covers an outer circumferential surface of the electric wire, the exterior tube comprising a plurality of recesses; a path restrictor that is attached to an outer circumferential surface of the exterior tube and restricts a path of the wire harness; and an attachment attached to an outer circumferential surface of a portion in a lengthwise direction of the path restrictor, wherein: the path restrictor includes a first main body that covers a portion of the outer circumferential surface of the exterior tube and an insertion port that is open in a direction orthogonal to a lengthwise direction of the first main body and extends over an entire length in the lengthwise direction of the first main body; the path restrictor has a connection connected to the attachment; the attachment has a cover that covers an outer circumferential surface of the connection; the cover comprises: a mark indicating a necessary minimum insertion amount of the path restrictor into the cover in a lengthwise direction of the wire harness main body; and a plurality of protrusions that fit respectively into the plurality of recesses of the exterior tube, each protrusion being separate from the mark; the minimum insertion amount mark is spaced from an end surface of the cover by a first distance; each protrusion among the plurality of protrusions is spaced from the end surface of the

cover by a distance shorter than the first distance; the minimum insertion amount mark is spaced from an end surface of the cover by a first distance; each protrusion among the plurality of protrusions is spaced from the end surface of the cover by a distance shorter than the first distance; the cover has a restricting wall that is engageable with a first end surface of the path restrictor; and the restricting wall is spaced from the end surface of the cover by a distance longer than the first distance.

11. A wire harness comprising: a wire harness main body including an electric wire and an exterior tube that covers an outer circumferential surface of the electric wire, the exterior tube comprising a plurality of recesses; a path restrictor that is attached to an outer circumferential surface of the exterior tube and restricts a path of the wire harness; and an attachment attached to an outer circumferential surface of a portion in a lengthwise direction of the path restrictor, wherein: the path restrictor includes a first main body that covers a portion of the outer circumferential surface of the exterior tube and an insertion port that is open in a direction orthogonal to a lengthwise direction of the first main body and extends over an entire length in the lengthwise direction of the first main body, the path restrictor has a connection connected to the attachment, the attachment has a cover that covers an outer circumferential surface of the connection, the cover comprises: a mark indicating a necessary minimum insertion amount of the path restrictor into the cover in a lengthwise direction of the wire harness main body; and a plurality of protrusions that fit respectively into the plurality of recesses of the exterior tube, each protrusion being separate from the mark, and the minimum insertion amount mark is recessed outward from an inner surface of the cover, in a radial direction of the cover, and does not pass through the cover in the radial direction of the cover.
